example: no 1:
from Book:

Use chain Rule to find the derivatives of $w = xy$ with respect to t along the Path $x = \cos t$, $y = \sin t$, what is the derivatives value at $t = \pi/2$.

→ From youtube
MC:
→ $f(x, y)$
$f(x, y, z)$
$f(x, y, z, t)$

Now the case?

Let's suppose
$Z = f(x, y)$ any function, means more than one variable.
is kai andar jo aagy x and y hain it further agr define hain like.

$g(t)$ $h(t)$

here: x function hai t ka aur y bi function hai t variable ka.

is kaa mtlb jo hamar function z hai
aur us z kai andar hamary pass khuch variables hain
aur jo variable s hai aagy just ek variables kai
andar define hain.

mean x, be t kai andar define hai
aur y be t main.

to agr ese function kaa aap sai
poochy

$\frac{dz}{dt} = ?$ derivative find out kary.

to is k liye hum chain rule use
kary gai.

How we will find?

$$\frac{dz}{dt} = \frac{df}{dx}\frac{dx}{dt} + \frac{df}{dy}\frac{dy}{dt}$$

it d.

→ means pehly jo humy function diya hai
us kaa humy partial derivative lai gai hao ek
variable kai according. then agr function x hai
to us kaa t kai acording derivative lai kar
partial kai saath multiply kro lai gai.

Note: in case i x and y are defined
in same variable t. so what if they
are defined in different?
this thing we will discuss in
case ii

example:-

compute: $\frac{dz}{dt}$ for $z = xe^{xy}$

$x = t^2$ and $y = t^{-1}$

→ so given Question is defined in the same variable hence we will use the case i

so first find.

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x}\left[xe^{xy}\right] \quad y = \text{constant treat}$$

so this is in product. apply productrule.

$$= x\frac{\partial}{\partial x}(e^{xy}) + e^{xy}\frac{\partial}{\partial x}(x)$$

$$= xe^{xy} \cdot y(1) + e^{xy} \cdot (1)$$

same because constant

its derivative of x.

$$= xe^{xy} \cdot y + e^{xy}.$$

$$= xye^{xy} + e^{xy}.$$

$$\frac{\partial z}{\partial x} = (xy+1)e^{xy} \rightarrow (i)$$

Now.

with respect to y

$$\frac{dz}{dy} = \frac{d}{dy}\left[ne^{ny}\right]$$ Now n = constant treat.

$$\boxed{n\frac{d}{dy}(e^{ny}) + e^{ny}\frac{d}{dy}(n) \quad ne^{ny} \quad n(1) + e^{ny}} \times$$

So here n will treat as a constant to is wajah sei ek hee term cosider hoga e^{ny}.

$$= n\frac{d}{dy} e^{ny} \cdot n(1)$$

sam

$$= n\frac{d}{dy} e^{ny} \cdot n$$

$$= n^2 e^{ny} \rightarrow (2)$$

Now we have to find value of $\frac{dn}{dt}$

$$\frac{dn}{dt} = \frac{d}{dt}(t^2) = 2t^{2-1}\frac{d}{dt}(t)$$

$$= 2t \rightarrow eq\ (3)$$

Now $$\frac{dy}{dt} = \frac{d}{dt}(t^{-1}) = -t^{-1-1} = -t^{-2}$$

or $$\boxed{-\frac{1}{t^2}} \rightarrow (4)$$ ~~Ans~~.

Now just replace it into the formula.

$$\frac{dz}{dt} = \frac{df}{dx} \times \frac{dx}{dt} + \frac{df}{dy} \times \frac{dy}{dt}$$

$$\frac{dz}{dt} = (xy+1)e^{xy} \times \left[2t\right] + x^2 e^{xy} \left[-\frac{1}{t^2}\right]$$